SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 2

ARMS PORTAL – POSITIVE TEST SCENARIOS AND TEST CASES

1. Aim:

To write a case study to identify and analyze the requirements of ARMS Portals, and prepare positive test scenarios as well as test cases for the application.

2. Problem Description

The ARMS Portal is an academic/administrative web portal through which authorized students and staff can sign in, view academic information, submit requests, upload documents and track request status. The case study assumes these common portal functions; institution-specific rules should be confirmed from the official ARMS specification.

The ARMS Portal is expected to provide reliable, secure and user-friendly services. The testing team verifies the stated requirements using valid and invalid inputs, records actual results, and reports any deviation from the expected behaviour.

2. Functional Requirements

1. Students and staff must sign in using valid institutional credentials.
2. The portal must display information only to the authenticated user with the appropriate role.
3. Students must be able to view their profile and academic details.
4. Users must be able to submit permitted requests with required information.
5. The portal must validate uploaded documents for file type and size.
6. Submitted requests must receive a unique reference number.
7. Users must be able to view the status of their own requests.
8. The portal must display clear validation and access-denied messages.

3. Positive Test Scenarios and Test Cases

Positive testing checks whether the application performs the required operation when valid data and correct user actions are supplied.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
PT01	Student login	Valid student ID and password	Student dashboard is displayed.	Pass
PT02	View profile	Open My Profile after login	Correct student profile is displayed.	Pass
PT03	View academic details	Select current semester	Registered courses and academic details are displayed.	Pass
PT04	Submit request	Complete all mandatory fields and submit a permitted request	Request is saved and a reference number is generated.	Pass
PT05	Upload valid document	Upload a permitted PDF within the size limit	Document is uploaded successfully.	Pass
PT06	Track request	Open a request using its reference number	Current request status is displayed.	Pass
PT07	Staff login	Valid staff credentials	Staff dashboard is displayed with staff permissions.	Pass
PT08	Logout	Select Logout	Session ends and login page is displayed.	Pass

5. Detailed Execution of a Failed Test Case

Test Case ID: PT08

Precondition: A registered user has a valid account and the required service is available.

Test Steps: 1. Sign in with valid credentials. 2. Open the relevant module. 3. Enter valid data. 4. Submit the operation.

Expected Result: The application completes the operation and displays a success confirmation.

Actual Result: The operation completes successfully in this illustrative execution.

Conclusion: The positive test passes when actual behaviour matches the expected result.

Conclusion

The test cases prepared for the ARMS Portal provide coverage of its key functional requirements. Executing them helps identify functional gaps, improve error handling and increase confidence in the quality of the application.